

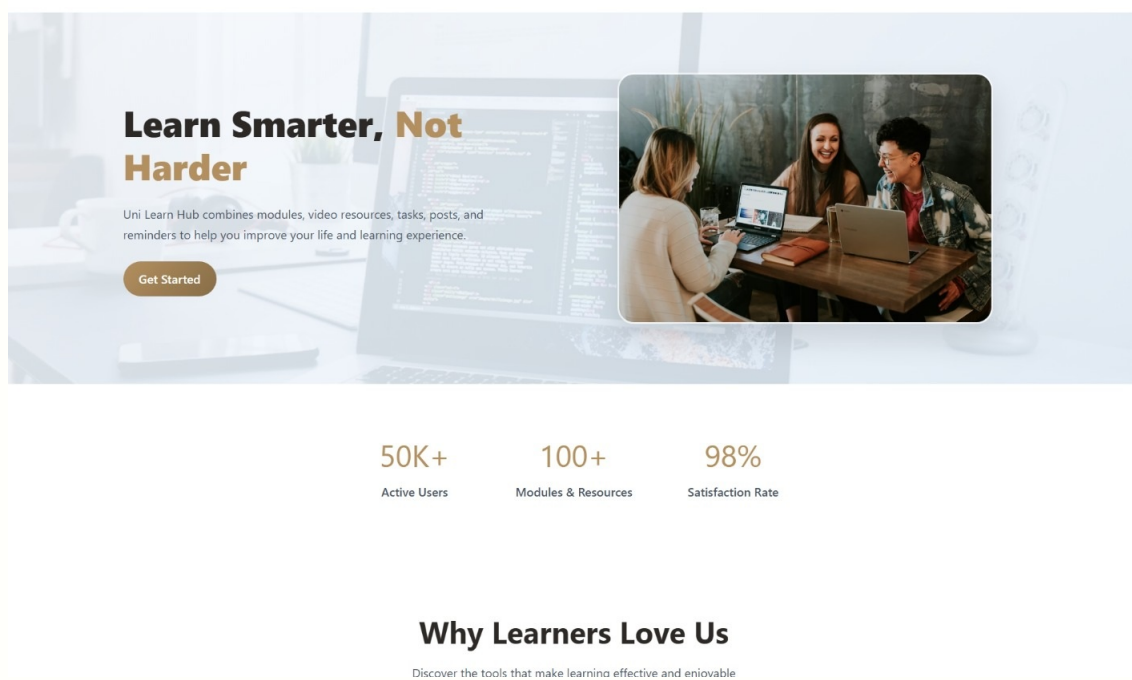


Uni Learn Hub

Full Stack Learning Platform

Professional project case study showcasing the completed Uni Learn Hub system, real UI screenshots, member contributions, and full-stack development work.

Project UI Preview



Spring Boot

React

PostgreSQL

JWT Auth

WebSocket

SSE

SMTP OTP

File Uploads



Project Snapshot

A complete learning platform that combines user accounts, resources, knowledge sharing, problem support, reviews, and gamified task progress.

Backend

**Spring Boot 4.0.3 /
Java 21**

Frontend

React 19 + Vite 8

Database

PostgreSQL

Auth

**Spring Security +
JWT cookies**

Realtime

WebSocket + SSE

Uploads

**Videos, PDFs,
images, chat files**

Real modules found in the codebase

Auth & User Management

Problem Asking / Chatbot

Realtime Support Chat

Website Review System

Resources Management

Knowledge Sharing Posts

Task Management & Rewards

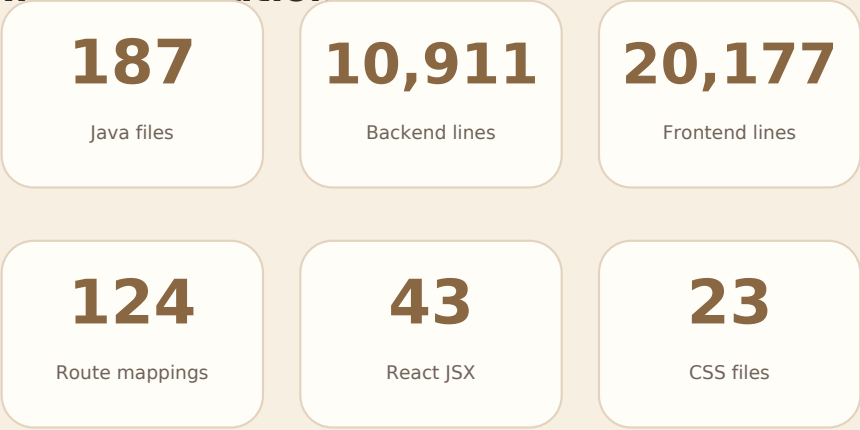
Quizzes & Learning



Technical Project Overview

A clean implementation summary based on the completed Uni Learn Hub system, covering backend, frontend, routes, modules, and security-focused documentation.

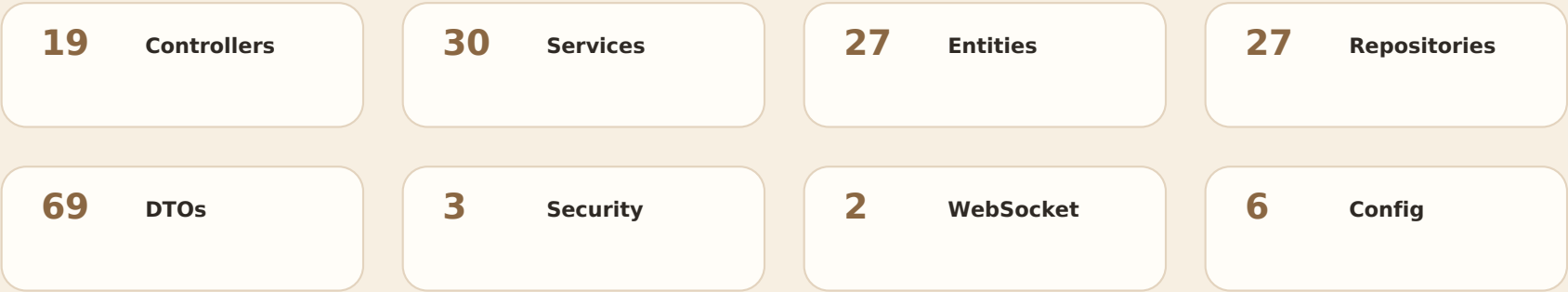
Implementation metrics



Architecture evidence

- **Backend API** Controllers, DTOs, services, repositories
- **Frontend UI** React pages, route views, reusable components
- **Database** PostgreSQL persistence with entity mapping
- **Realtime** WebSocket/SSE flow for live chat and alerts

Backend structure



PUBLIC DOCUMENTATION NOTE

This portfolio PDF highlights architecture, implementation modules, and UI evidence while excluding private environment values such as local database credentials, SMTP keys, and JWT secrets.

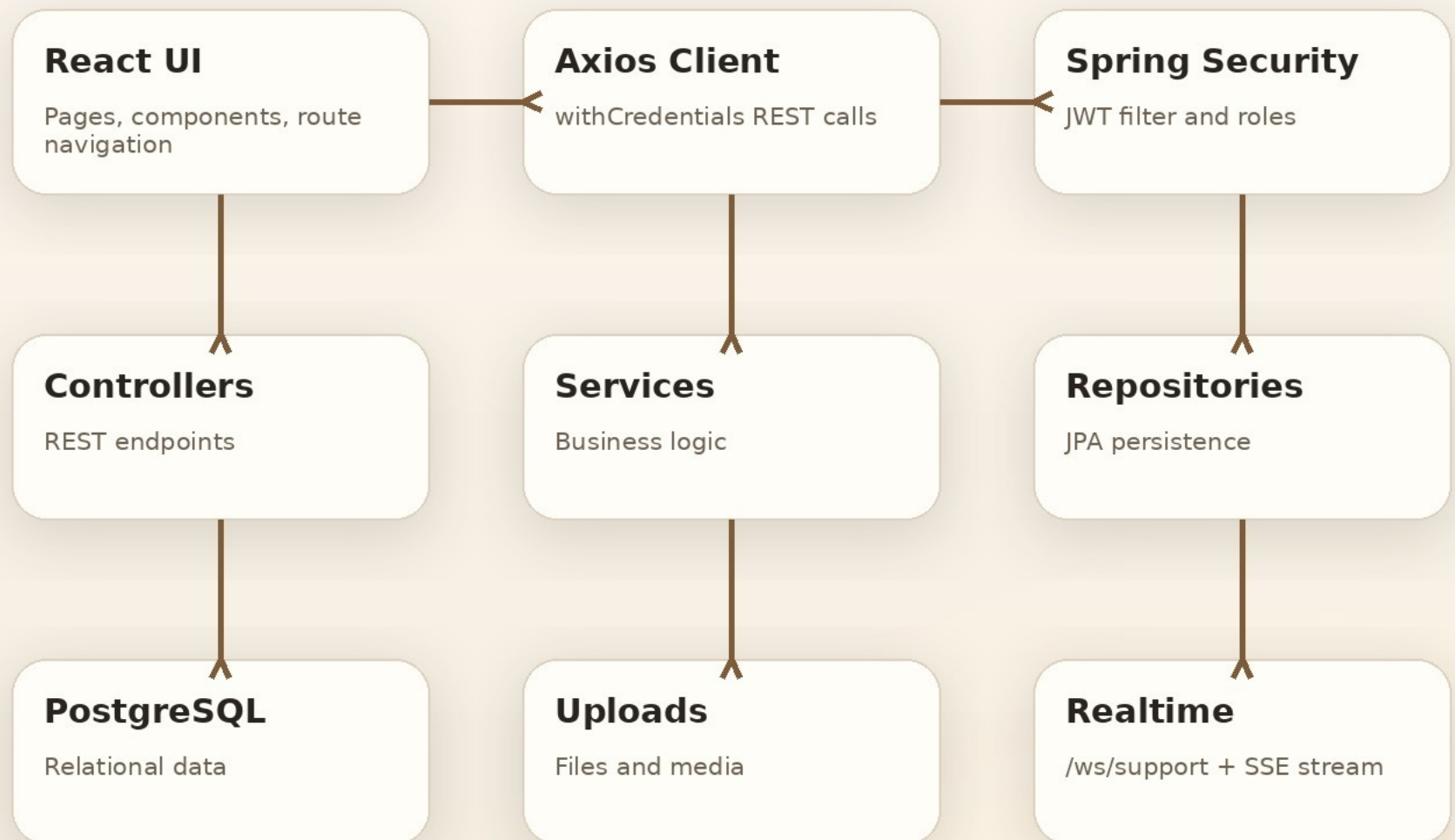
System modules reviewed

- User Management
- Resources Management
- Knowledge Sharing
- Task Management
- Realtime Chat
- Reviews
- Problem Support
- Rewards & Progress



Clean System Architecture

React provides the learner/admin interface. Spring Boot handles API security, business logic, realtime updates, persistence, and file handling.





Member-wise Contribution Map

Screenshots are grouped directly under each member name so the report clearly shows ownership and evidence.



Anuradha Dolomulla

User Management, problem asking, realtime support chat, user-to-user and user-to-admin chat, review system.



Yohan Kavinda

Resources management, admin resource control, shared library, video/PDF/notes uploads and downloads.



Sasindi Bhagya

Knowledge sharing, content management, community stream, posts, actions, saved collections, alerts.



Laksara Manodya

Task management, admin assignment, user progress, XP rewards, badges, and game unlocks.



Anuradha Dolomulla

User Management, Problem Asking, Realtime Chat and Review System

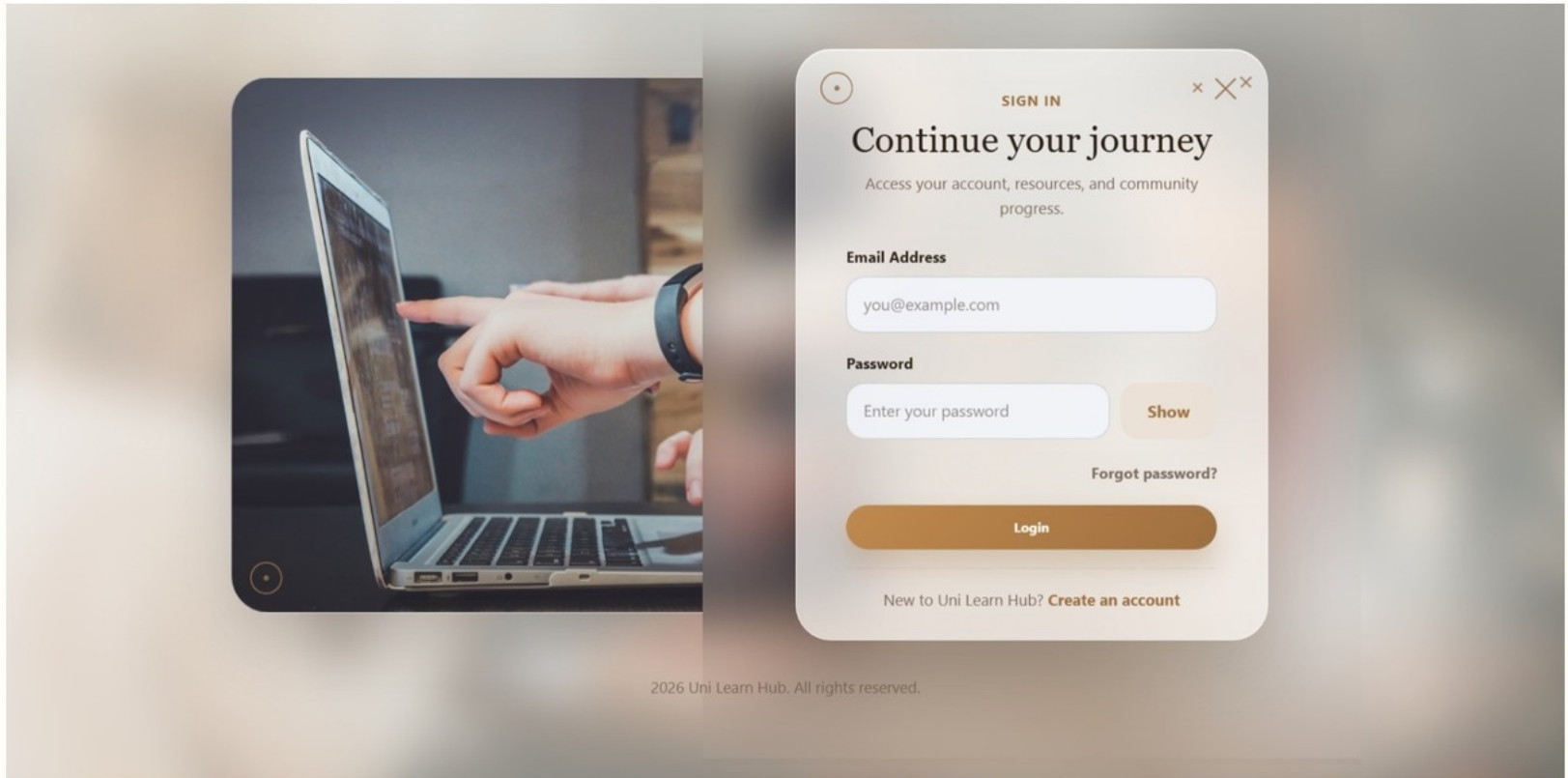
- Authentication, registration, OTP verification, forgot password, profile settings, and account security.
- Problem asking assistant and realtime support messenger for user-to-user and user-to-admin communication.
- Website review system where learners submit ratings, sentiment, and public testimonials.



Login Page - User Access Gateway

Professional sign-in screen connected to the backend authentication flow.

ANURADHA DOLOMULLA



Module: User Management / Authentication

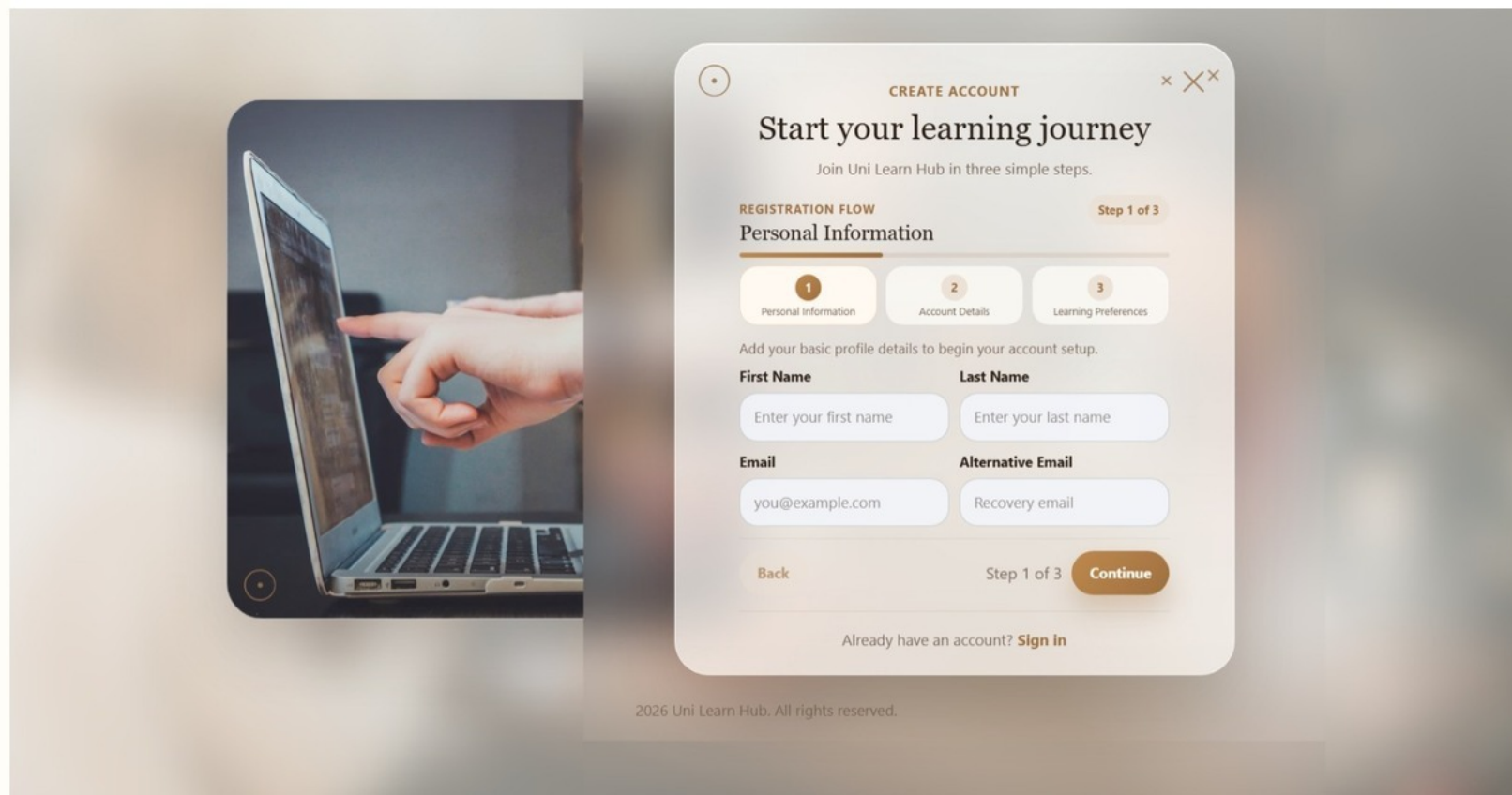
- Email/password login page with polished Uni Learn Hub theme.
- Frontend uses Axios with credentials enabled for secure cookie-based sessions.



Registration Flow - Step 1

Personal information step of the three-step account creation journey.

ANURADHA DOLOMULLA



Module: User Registration Flow

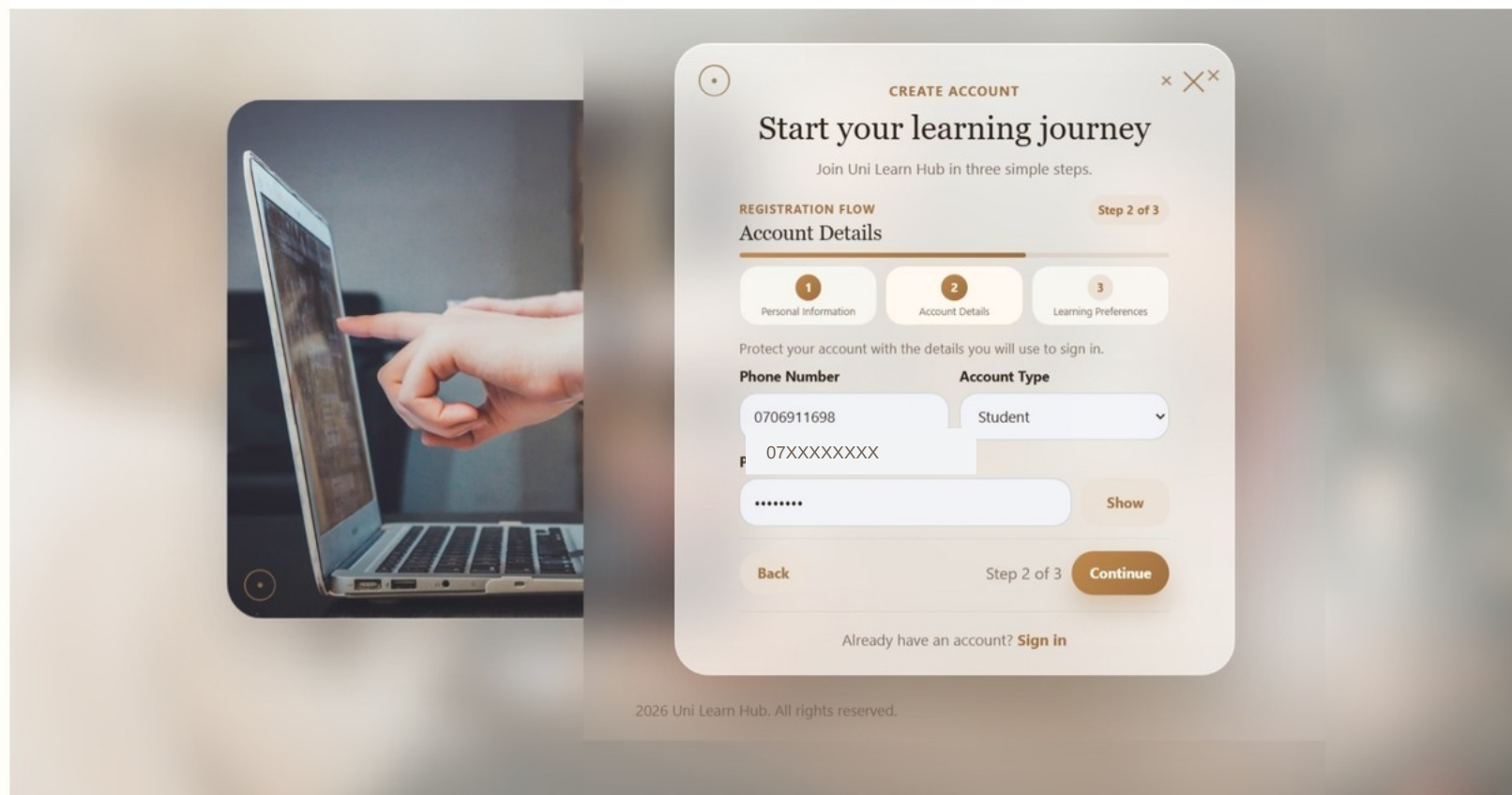
- Collects first name, last name, email, and alternative recovery email.
- Clear progress indicator improves user onboarding experience.



Registration Flow - Step 2

Account details step with phone, role/account type, and password setup.

ANURADHA DOLOMULLA



Module: User Registration Flow

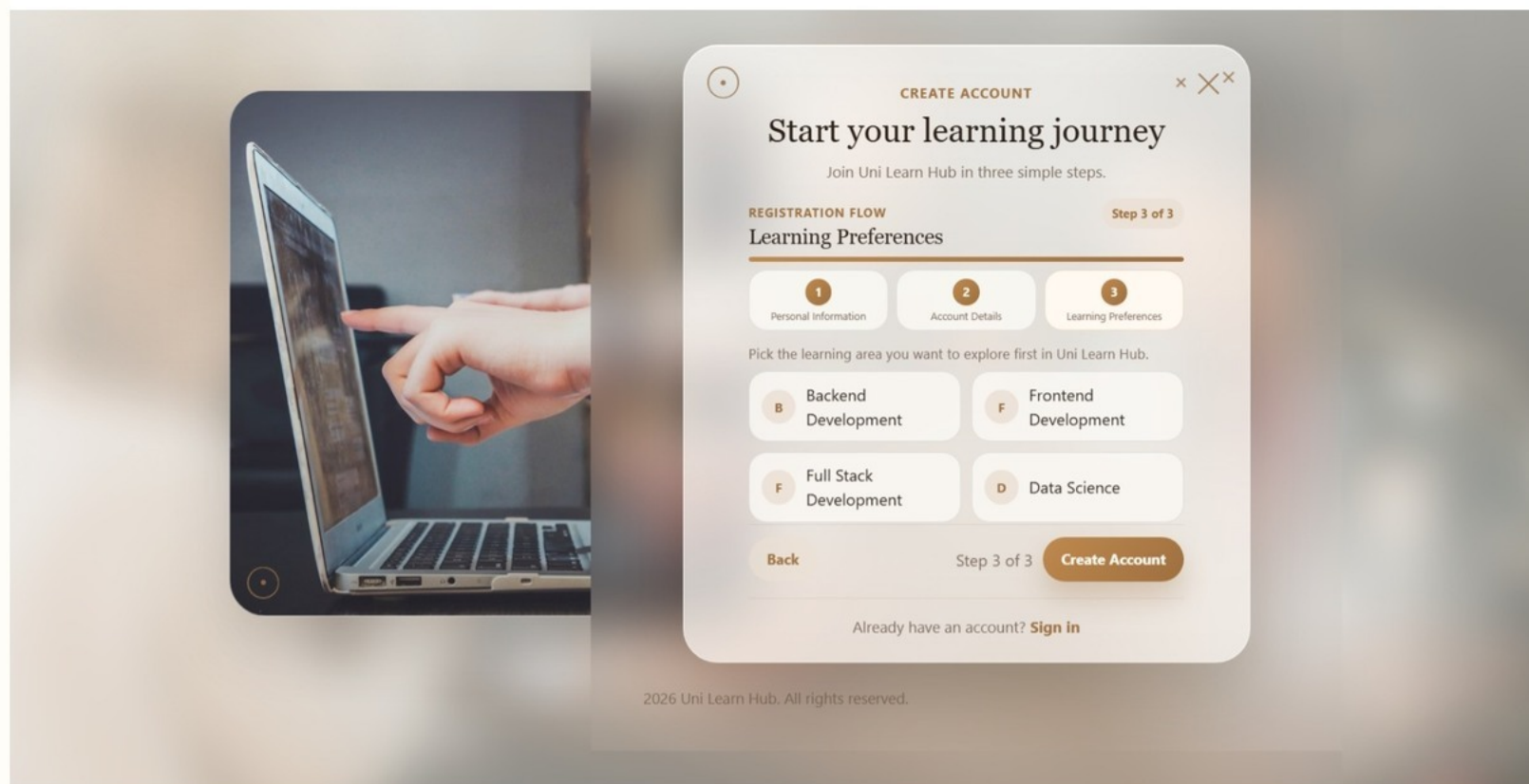
- Captures phone number and student/admin account type.
- Password field supports show/hide interaction and guided flow.



Registration Flow - Step 3

Learning preference selection completes the registration journey.

ANURADHA DOLOMULLA



Module: User Registration Flow

- Learners select initial area such as Backend, Frontend, Full Stack, or Data Science.
- Preference can be used for personalised content and alerts.

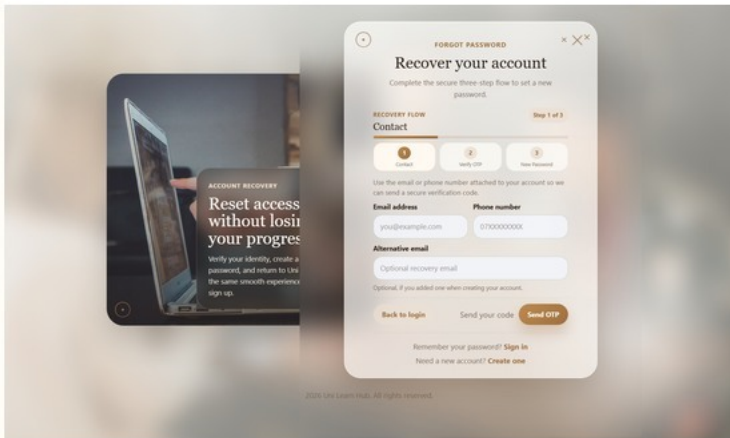


Forgot Password and OTP Verification

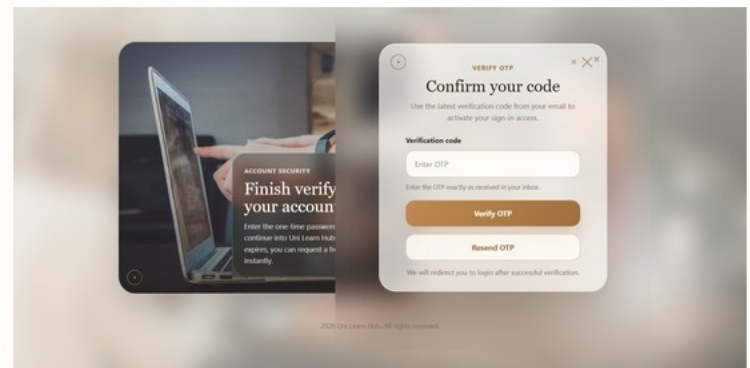
Account recovery and email verification screens support secure user lifecycle management.

ANURADHA DOLOMULLA

Recover account



Verify OTP



Module: Forgot Password + OTP Verification

- Recovery checks multiple identifiers before sending an OTP.
- OTP verification issues a short-lived VERIFY token before password reset.
- Resend limits reduce abuse during account recovery.



Account Settings and Profile Management

User profile settings page for name, email OTP update, password change, image updates, and danger-zone deletion.

ANURADHA DOLOMULLA

 **Account Settings**

[← Back to Profile](#)



anuradha dolomulla
anuradha333dh@gmail.com
anuradha@example.com

[Change Cover](#)

 **Update Name**

FIRST NAME

LAST NAME

 **Update Email**

NEW EMAIL ADDRESS

VERIFICATION CODE

 **Change Password**

 **Danger Zone**

Module: User Profile / Account Settings

- Profile and cover image update support.
- Name, email verification, password change, and secure account delete area.



Problem Asking System

Landing chatbot and assistant panel allow learners to ask common learning questions.

ANURADHA DOLOMULLA

Landing assistant



50K+ Active Users 100+ Modules & Resources 98% Satisfaction Rate

Why Learners Love Us

Discover the tools that make learning effective and enjoyable

- 01 Modules & Videos
Access video resources and modules to learn efficiently.
- 02 Tasks & Achievements
Create tasks, achieve goals, and improve your productivity.
- 03 Reminders
Set reminders to stay on track and improve your life.
- 04 Posts & Questions
Ask questions, post updates, and interact with other users.

Chat panel

ASK THE CHATBOT

Online

X

Uni Learn Assistant

What is Uni Learn Hub?

How to learn step by step?

What is deep learning?

YOU

What is Uni Learn Hub?

ASSISTANT

Uni Learn Hub is an online platform for university students to share knowledge and

Ask here...

Send

Module: Problem Asking / Chatbot

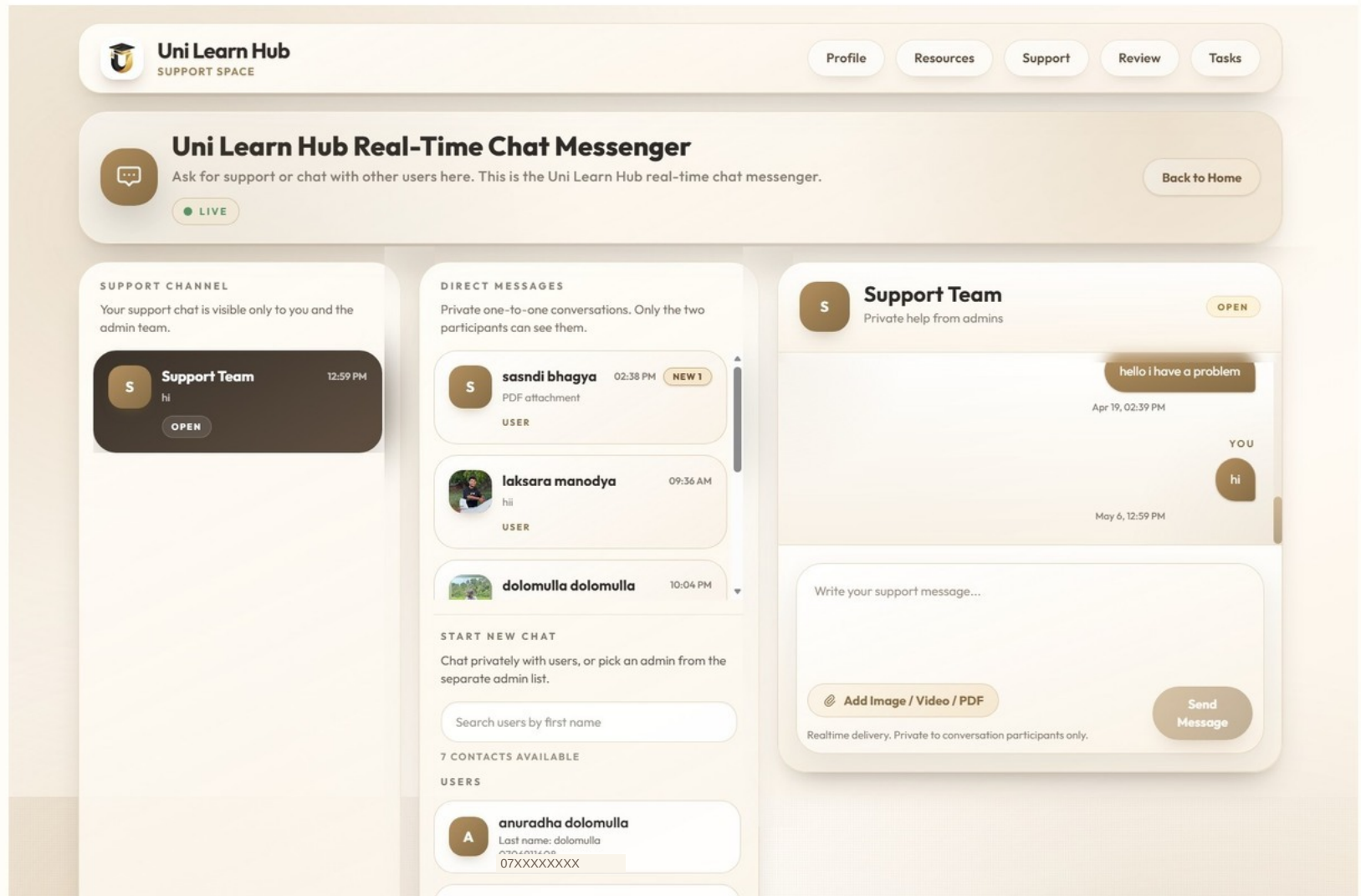
- Concept assistant answers common learning questions from stored topics.
- Backend /api/chat uses fuzzy matching for simple support responses.
- Landing chatbot keeps problem asking visible before login.



Realtime Support Messenger

Support page for realtime user-admin support and direct user-to-user conversations.

ANURADHA DOLOMULLA



Module: Realtime Chat / Support System

- Support Team thread for admin communication.
- Direct messages between users with media/file attachment option.
- Backend includes /support APIs and /ws/support WebSocket channel.

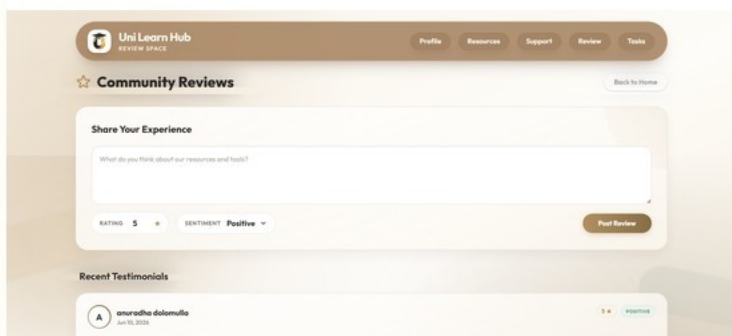


Website Review System

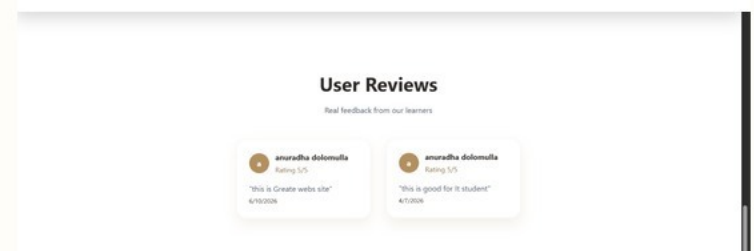
Learners can post reviews and public testimonials are displayed on the website.

ANURADHA DOLOMULLA

Review workspace



Public cards



Module: Review System

- Users create ratings with positive, neutral, or negative sentiment.
- Users can update and delete only their own reviews.
- Public review cards show testimonials from submitted feedback.



Yohan Kavinda

Resources Management

- Learner shared resource library for videos, PDFs, and notes with filters and downloads.
- Admin resource control center for instructor-uploaded video materials.
- Spring Boot resource APIs for upload, stream, filter, download, update, and delete flows.



Shared Resource Library - Top View

Learner-facing resource center with upload form, accepted file rules, counters, and filters.

YOHAN KAVINDA

**Uni Learn Hub**
RESOURCE CENTER

[Profile](#)[Resources](#)[Support](#)[Review](#)[Tasks](#)

**Shared Resource Library**

This page now shows only user shared videos, PDFs, and notes. Instructor uploaded learning videos are available on a separate page.

[View Instructor Resources](#)[Back to Home](#)

Community Shared Resources

Videos, PDFs, and notes are arranged in separate spaces now, while the current upload and download flow stays the same.

Accepted files
Video: MP4, MOV, AVI, MKV, WEBM
Documents: PDF, DOC, DOCX, TXT, RTF, MD, PPT, PPTX

Share Your Resource

FILE

[Choose File](#) No file chosen

Resource title

Short description

Description must be at least 20 characters. Current: 0

Select year

Select semester

[Share Resource](#)

Shared by Users

5 resources in this view

VIDEOS
1

PDFS
1

NOTES
3

All Years

All Semesters

All Types

Search shared resources...

Search

Reset

Video Gallery

1 ITEM

Play shared lessons, walkthroughs, and recorded explainers.

Introduction to Programming for Beginners

Uploaded by anuradha dolomulla (USER)

VIDEO

Programming is the process of giving instructions to a computer to perform specific tasks.



Module: Resources Management - User Side

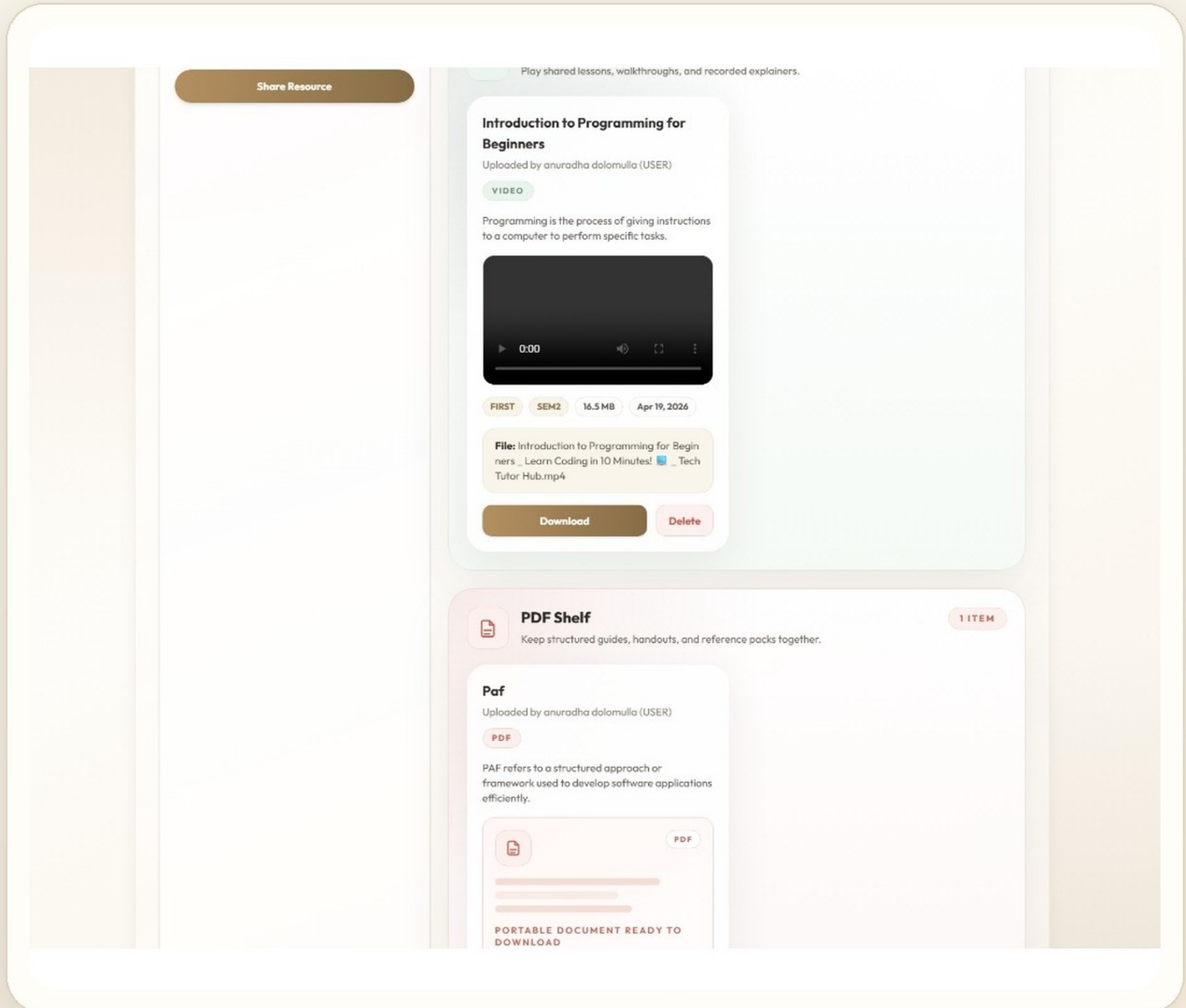
- Separate user shared resources from instructor resources.
- Supports video, PDF, notes and structured search filters.



Shared Resource Library - Video and PDF Area

Resource content is split into gallery and shelf areas for easier learning material discovery.

YOHAN KAVINDA



Module: Resources Management - User Side

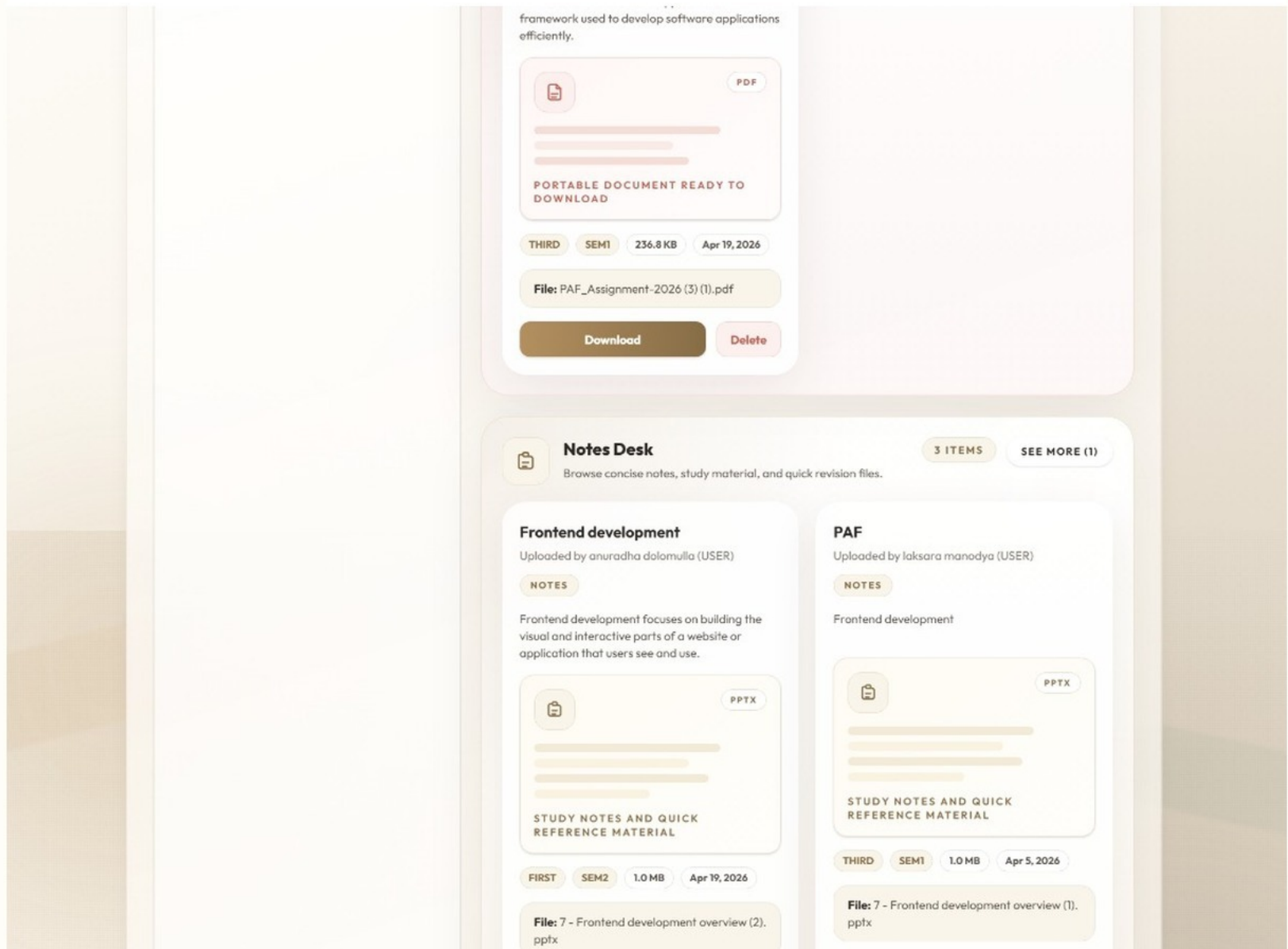
- Video gallery section shows media preview and download action.
- PDF shelf section keeps handouts and reference packs organised.



Shared Resource Library - Notes Desk

Notes desk presents quick study files and reference material as separate learning resources.

YOHAN KAVINDA



Module: Resources Management - User Side

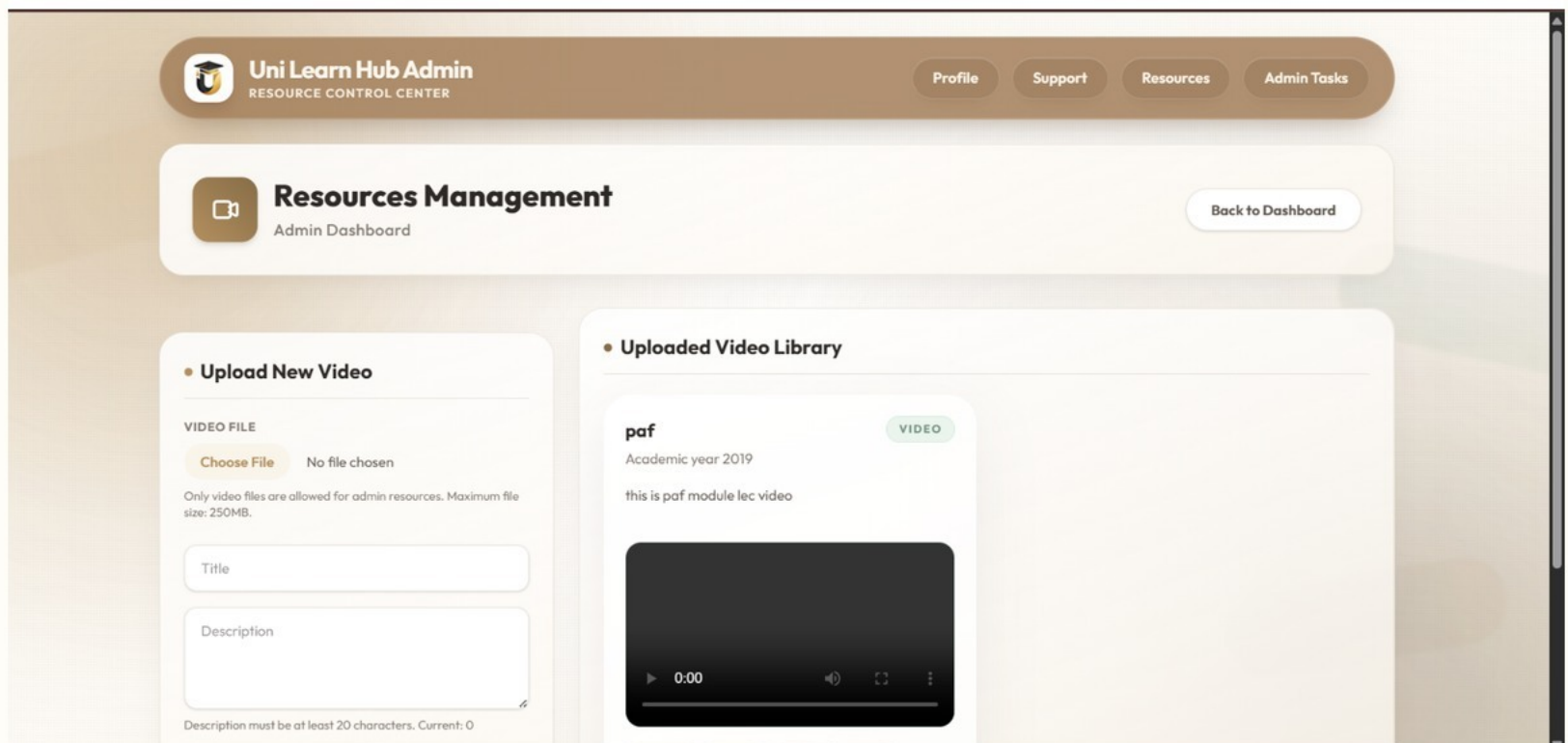
- Notes are separated from videos and PDFs for cleaner browsing.
- Download and delete actions support resource lifecycle management.



Admin Resource Control Center

Admin dashboard for uploading new videos and managing the instructor video library.

YOHAN KAVINDA



Module: Resources Management - Admin Side

- Upload new video with title and description.
- Admin can view and control uploaded video resources.



Sasindi Bhagya

Knowledge Sharing and Content Management

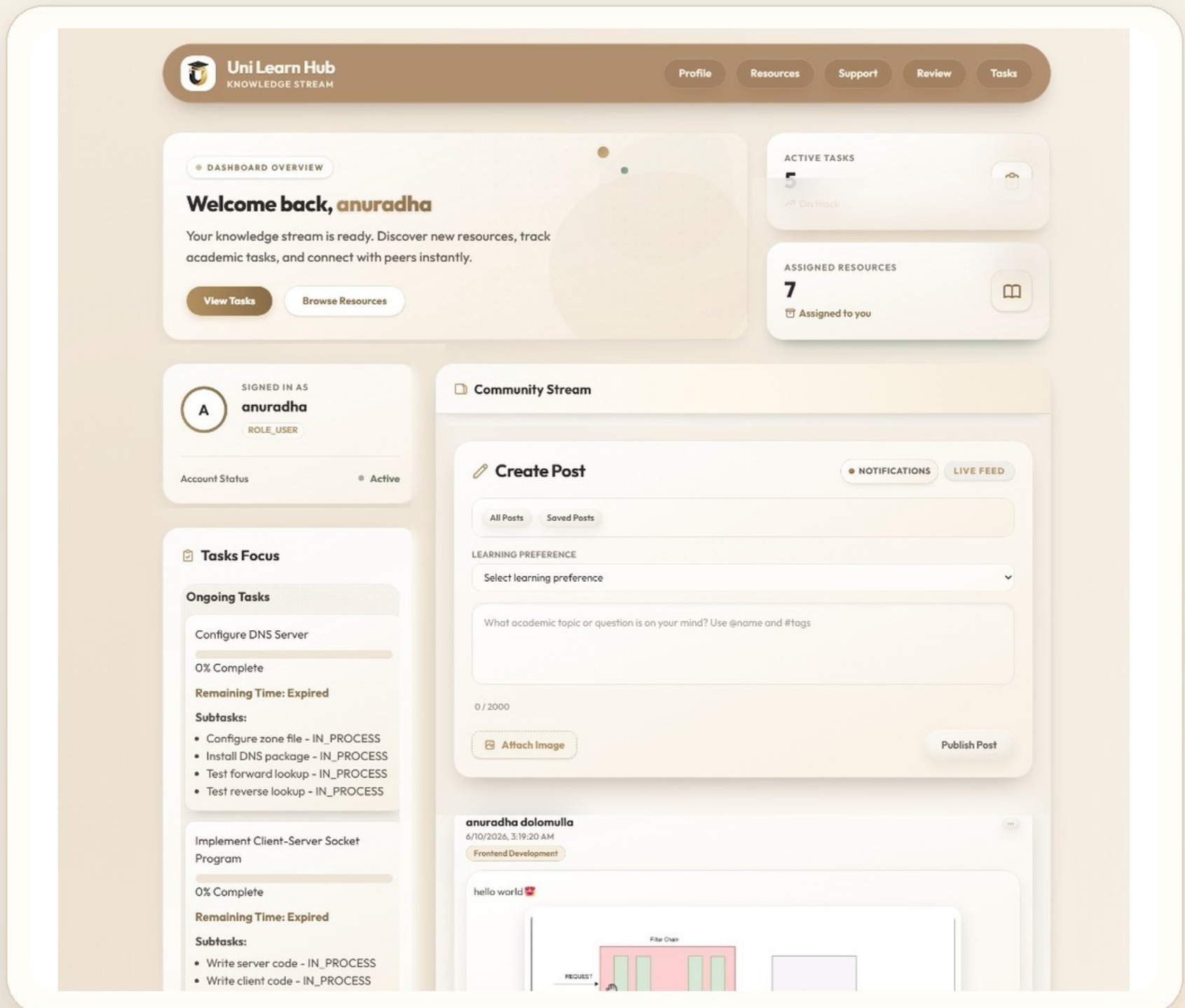
- Community stream with post creation, learning preference categories, images, hashtags, and mentions.
- Content actions such as save, share, delete, comments, reactions, and saved collections.
- Preference alerts and live-feed notifications for newly published learning content.



Knowledge Stream Dashboard

Community stream dashboard connects tasks, resources, account status, and academic posts in one learner workspace.

SASINDI BHAGYA



Module: Knowledge Sharing / Content Management

- Create post form with learning preference dropdown and image attachment.
- Feed shows learner posts with comments, likes, shares, and attachments.



Post Creation and Feed Actions

Post page supports saved posts, collections, delete options, shares, reactions, and comments.

SASINDI BHAGYA

The mockup displays a post creation interface at the top with tabs for 'All Posts' and 'Saved Posts', and filters for 'All Collections', 'coll (1)', 'Saved (1)', and 'ss (0)'. Below these is a 'LEARNING PREFERENCE' dropdown menu. A text input field prompts the user with 'What academic topic or question is on your mind? Use @name and #tags', with a character count '0 / 2000'. An 'Attach Image' button and a 'Publish Post' button are at the bottom of the form.

Below the form, a post by 'anuradha dolomulla' is shown, dated '6/10/2026, 3:19:20 AM' with the tag 'Frontend Development'. The post content is 'hello world 🍷'. To the right of the post is a menu with 'Delete post', 'Saved', and 'Share' options.

The post content includes a diagram titled 'Filter Chain' illustrating the request processing flow:

```
graph LR
    REQUEST --> FC[Filter Chain]
    FC -- RESPONSE --> FC
    FC --> AC[Your Application Controllers]
    AC -- matches() --> PE[PasswordEncoder]
    AP[Authentication Provider] --> PE
```

The diagram shows a 'Filter Chain' box with four vertical bars. A 'REQUEST' arrow enters from the left, and a 'RESPONSE' arrow exits to the left. An arrow points from the 'Filter Chain' to 'Your Application Controllers'. From there, an arrow labeled 'matches()' points to a 'PasswordEncoder' box. An 'Authentication Provider' box also has an arrow pointing to the 'PasswordEncoder'.

Module: Knowledge Sharing / Post Management

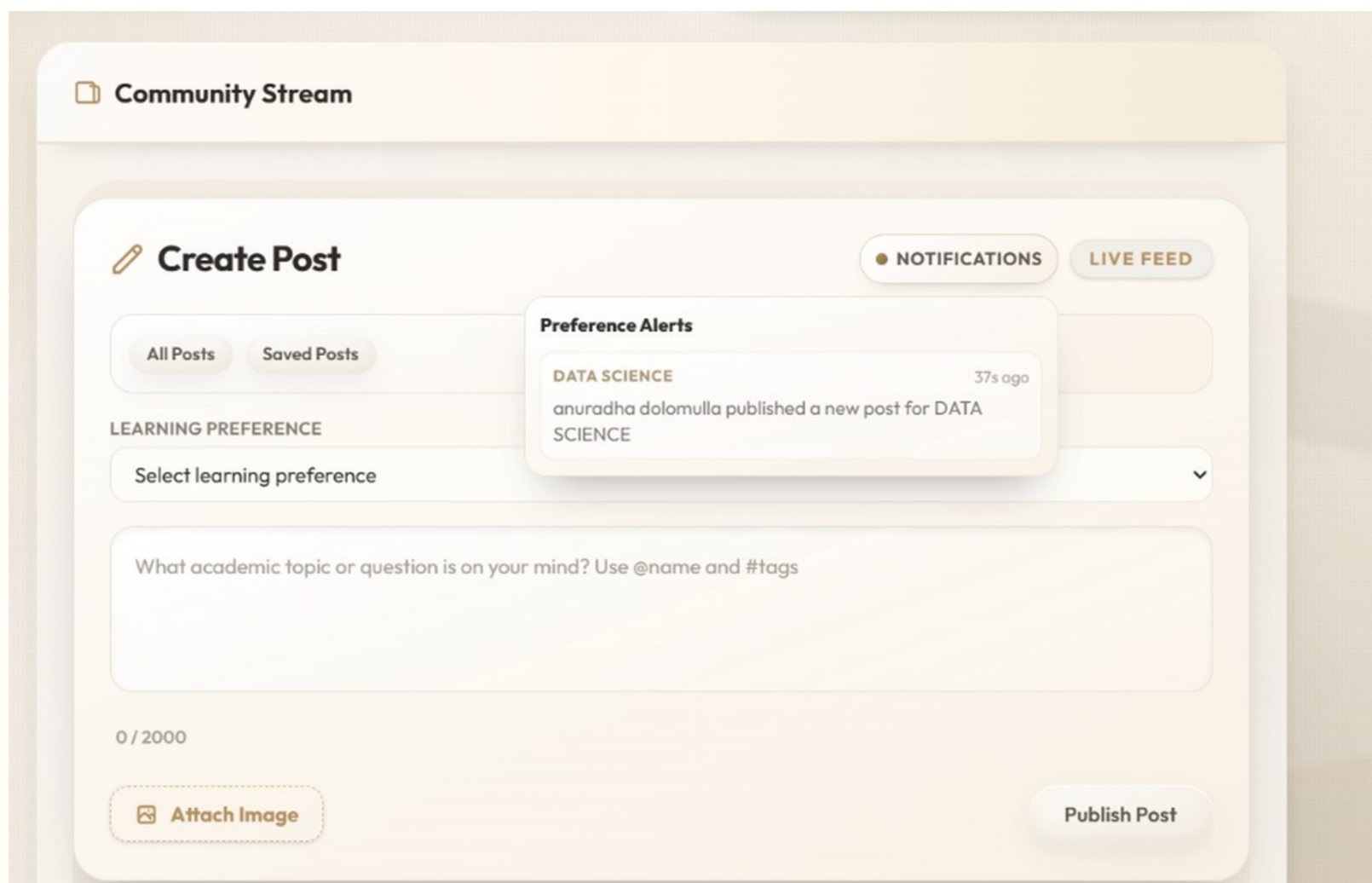
- Post composer supports text content, tags and image attachment.
- Post menu supports delete, saved and share actions.



Preference Alerts and Live Feed

Notification popover alerts users when new content is published for selected learning preferences.

SASINDI BHAGYA



Module: Knowledge Sharing / Notifications

- Preference-based alerts improve content discovery.
- Live feed indicator shows active update experience.



Laksara Manodya

Task Management - Admin and User Side

- Admin assigns structured learning tasks with deadlines, XP rewards, difficulty, learner selection, and subtasks.
- Learners complete assigned work, update subtask status, and track progress.
- Reward system adds XP, badges, levels, and game unlocks after task completion.



Admin Task Assignment Page

Admin creates and manages assigned tasks from a structured task manager dashboard.

LAKSARA MANODYA

Uni Learn Hub Admin

TASK MANAGER

Profile

Support

Resources

Dashboard

ADMIN CREATE TASK

Create and manage assigned tasks

Assign work to learners, review active admin tasks, and remove assignments cleanly from one place.

Back to Dashboard

Refresh Tasks

LEARNERS

6

Available for assignment

ASSIGNED TASKS

5

Admin-created tasks in the system

ADMIN PANEL

Assign learning tasks

Uni Learn Hub

Create structured work with reward-ready XP and deadlines.

TASK TITLE

Prepare database normalization notes

DESCRIPTION

Explain the learning goal and what "done" looks like.

Minimum 20 characters. Current: 0

DIFFICULTY

Medium

DEADLINE

mm/dd/yyyy

XP REWARD

75

ASSIGN USER

Select a learner

SUBTASKS

Write one subtask per line
Draft notes
Create examples
Review submission

Assign Task

ADMIN VIEW

Admin assigned tasks

5 tasks

Everything the admin has assigned across learners.

HARD

+200 XP

Completed

Delete

Implement Client-Server Socket Program 2

Develop a TCP client-server program in C. Task is done when client and server communicate successfully.

START

Apr 19, 2026

DEADLINE

Apr 20, 2026

PROGRESS

100%

ASSIGNED TO

anuradha dolomulla

write notes

Complete

COMPLETE

watch videos

Complete

COMPLETE

MEDIUM

+400 XP

In progress

Delete

Implement Client-Server Socket Program

Develop a TCP client-server program in C. Task is done when client and server communicate successfully.

START

Apr 19, 2026

DEADLINE

Apr 29, 2026

PROGRESS

0%

ASSIGNED TO

anuradha dolomulla

Module: Task Management - Admin Side

- Create task form includes title, description, difficulty, deadline, XP and assignee.
- Admin view displays active assigned tasks and learner counts.

Uni Learn Hub - Professional Project Case Study

26



Admin Assigned Task Review

Admin can review progress, subtask status, assignee, difficulty, XP, deadline, and delete actions.

LAKSARA MANODYA

Uni Learn Hub

ADMIN VIEW

Admin assigned tasks

5 tasks

Everything the admin has assigned across learners.

HARD

+200 XP

Completed

Implement Client-Server Socket Program 2

Develop a TCP client-server program in C. Task is done when client and server communicate successfully.

Delete

START

Apr 19, 2026

DEADLINE

Apr 20, 2026

PROGRESS

100%

ASSIGNED TO

anuradha dolomulla

write notes

Complete

COMPLETE

watch videos

Complete

COMPLETE

MEDIUM

+400 XP

In progress

Implement Client-Server Socket Program

Develop a TCP client-server program in C. Task is done when client and server communicate successfully.

Delete

START

Apr 19, 2026

DEADLINE

Apr 29, 2026

PROGRESS

0%

ASSIGNED TO

anuradha dolomulla

Write server code

In process

IN_PROCESS

Write client code

In process

IN_PROCESS

Module: Task Management - Admin Side

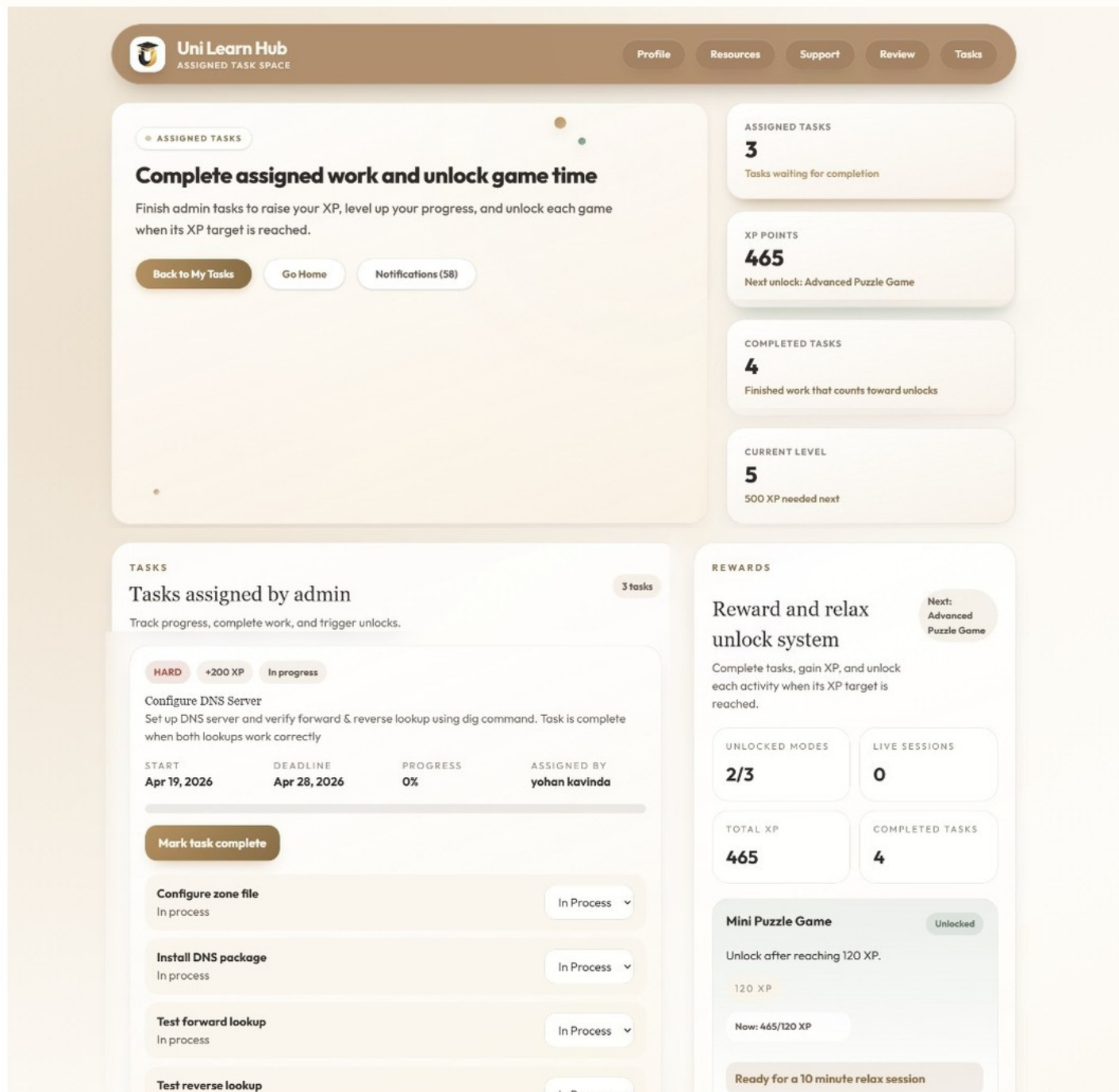
- Each assigned task card shows progress and subtask completion state.
- Admin can remove assignments when needed.



User Assigned Task Workspace

Learners view their assigned tasks, progress bars, subtasks, status dropdowns, and completion action.

LAKSARA MANODYA



Module: Task Management - User Side

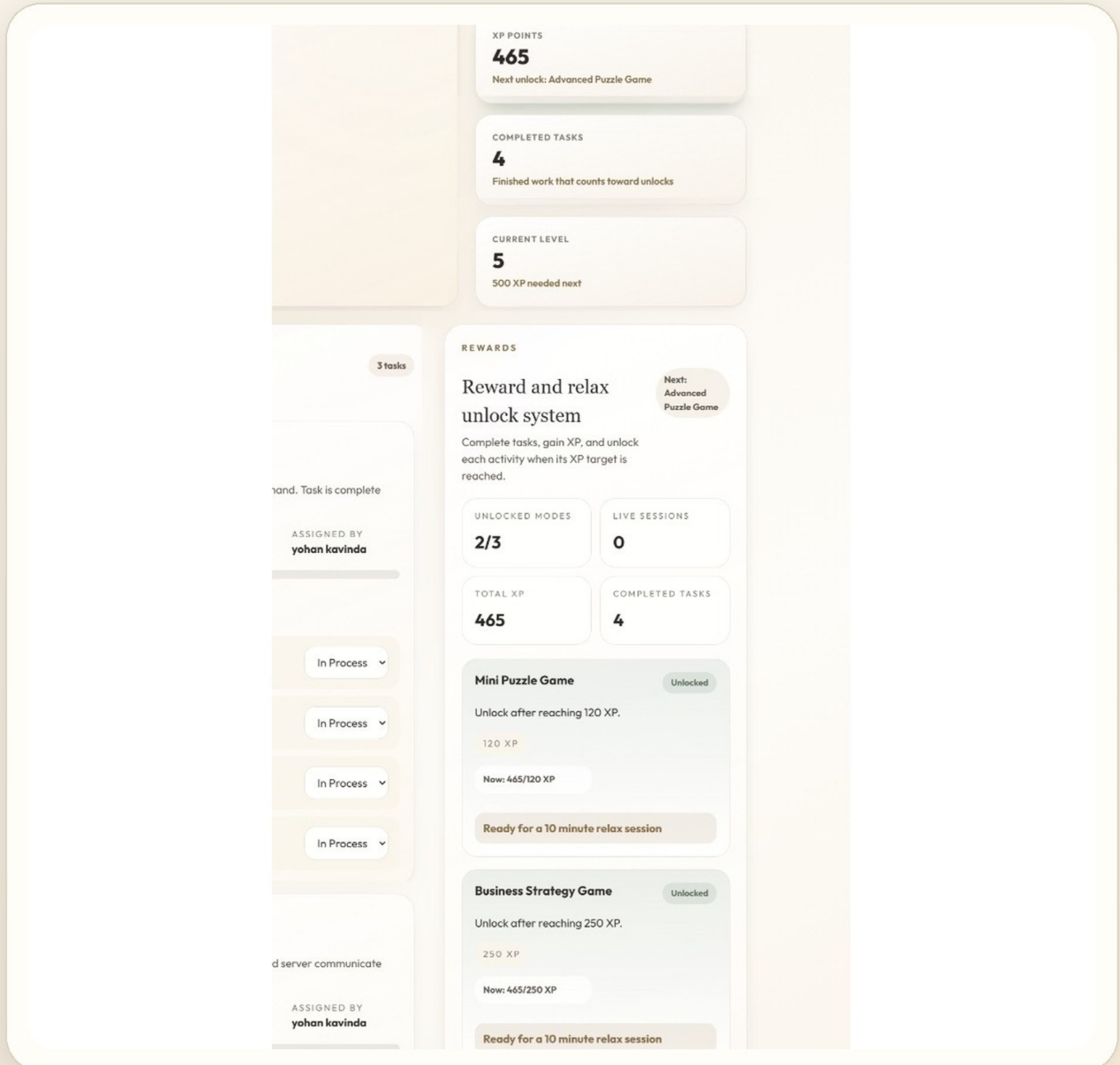
- Learners can mark tasks complete after finishing subtasks.
- Progress percentage is displayed clearly per assigned task.



Rewards and Gamified Progress

Task completion unlocks XP, badges, and learning-break games such as puzzle and business simulation.

LAKSARA MANODYA



Module: Task Management - Rewards / Gamification

- Reward panel shows XP, level, completed tasks, and unlock status.
- Mini games encourage task completion and learner engagement.



Backend API Coverage

The Spring Boot backend exposes module-based REST endpoints and separates controller, service, repository, entity, and DTO layers.

Authentication

/auth/register, /auth/login, /auth/verify-code, /forgotpass/*

Users & Admin

/user/me, /user/update-name, /user/update-email, /admin/dashboard

Support Chat

/support/bootstrap, /support/contacts, /support/direct-conversations, /ws/support

Reviews

/reviews, /reviews/gets, /reviews/{id} update/delete

Resources

/api/videos/*, /api/shared-resources/*, /quizzes/*

Posts

/posts/feed, /comments/*, /reactions/*, /shares/*, /saved-posts/*

Notifications

/notifications/my, /notifications/unread-count, /notifications/stream

Tasks

/tasks/my, /tasks/admin/assign, /tasks/games/*, /tasks/dashboard



Entity and Persistence Design

JPA entities and repositories support user accounts, resources, posts, chat, reviews, and task progress.

User Management

User

ForgotPassword

Role

Interest

Token

Resources & Quiz

VideoResource

SharedLibraryResource

QuizQuestion

UserVideoQuizAssignment

Knowledge Sharing

Post

Comment

Reaction

Share

SavedPost

PostCollection

Notification

Mention

Problem / Review

ChatConversation

ChatMessage

SupportMessage

SupportQuestion

Review

Concept

Task Management

MainTask

SubTask

Progress

Reward

GameUnlock

Notifications

Database is configured for PostgreSQL and uses Hibernate/JPA during development. Sensitive values from the ZIP are intentionally excluded from this PDF.

Database is configured for PostgreSQL and uses Hibernate/JPA during development. Sensitive values are intentionally excluded from this public documentation.



Interview Talking Points

A professional summary of what can be explained confidently from the real project codebase and UI evidence.

Full-stack architecture

React UI, Axios API client, Spring Boot controllers/services/repositories, PostgreSQL persistence.

Authentication flow

Registration, OTP verification, login, JWT cookies, Spring Security filter chain, role-based access.

Realtime support

/ws/support WebSocket sessions, direct/support conversations, admin routing, and HTTP message actions.

Content & resources

File uploads, streaming/download endpoints, posts, comments, saved posts, notifications, and filters.

Task gamification

Admin assignment, subtask progress, XP reward, badges, levels, unlock rules, and learner reward panel.

Security awareness

Secrets are redacted, and production deployment should use environment variables and HTTPS secure cookies.



Uni Learn Hub

Full Stack Learning Platform Case Study

Built as a group university project with Spring Boot, React, PostgreSQL, JWT security, realtime support chat, knowledge sharing, resources management, reviews, and task gamification.

Portfolio Ready

Interview Ready

University Project

Clean UI Evidence

Thank You

Prepared by the Uni Learn Hub project team